VIDEO GAME SALES

ANALYSIS REPORT

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REPORT OUTLINE

Dataset

Data Cleaning

Data Visualization

A/B Testing

Conclusion

DATASET

Video Game Sales

The dataset contains a list of video games with sales greater than 100,000 copies

DATA DICTIONARY

Sales in Millions

NA_Sales - Sales in North America

EU_Sales - Sales in Europe

JP_Sales - Sales in Japan

Other_Sales - Sales in Rest of the World

Global_Sales - Total Worldwide Sales

'**Misc**' Genre

diverse category that encompasses games that don't fit neatly into specific genres

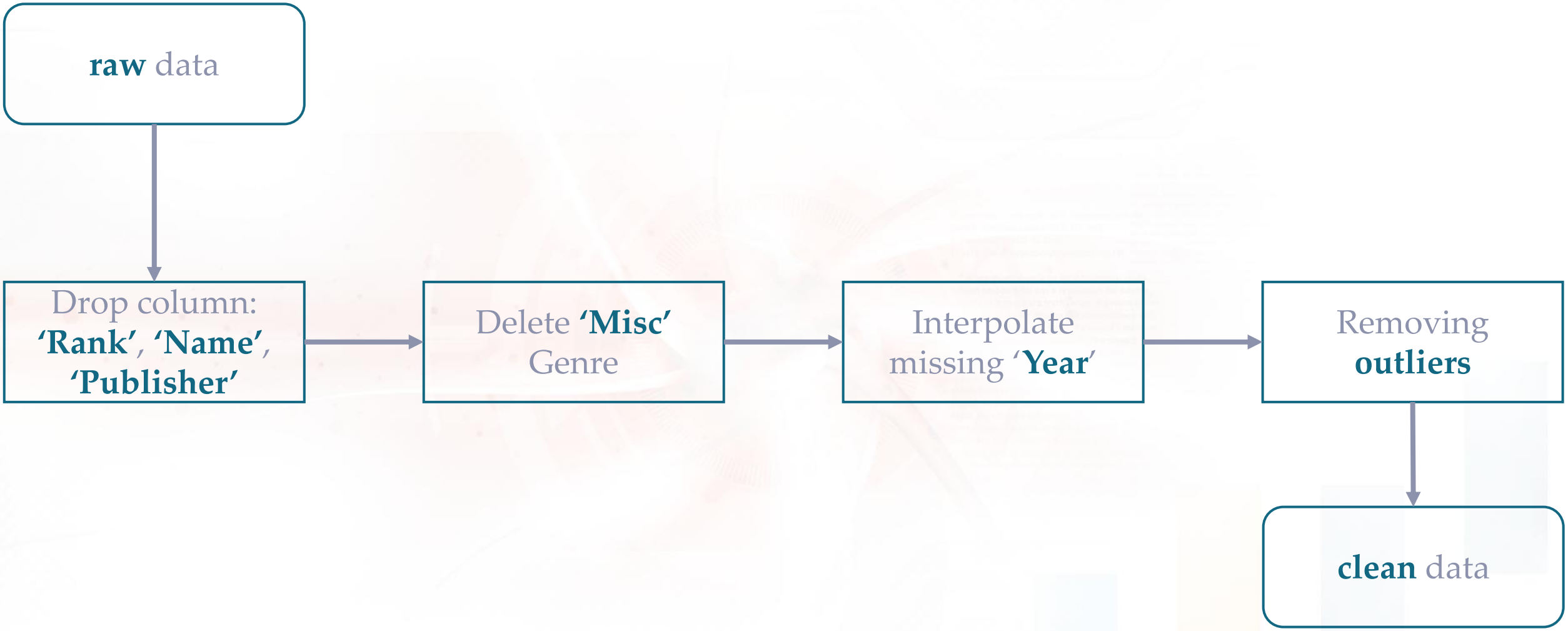
'**Action**' Genre

Games that emphasize physical challenges, hand-eye coordination, and reaction time.

e.g. platform, shooter, beat'em ups, and more

DATA CLEANING

DATA CLEANING STEPS



DATA VISUALIZATION

KEY QUESTIONS

What is the **top performing platform** by global sales?

in Europe, Japan, and North America?

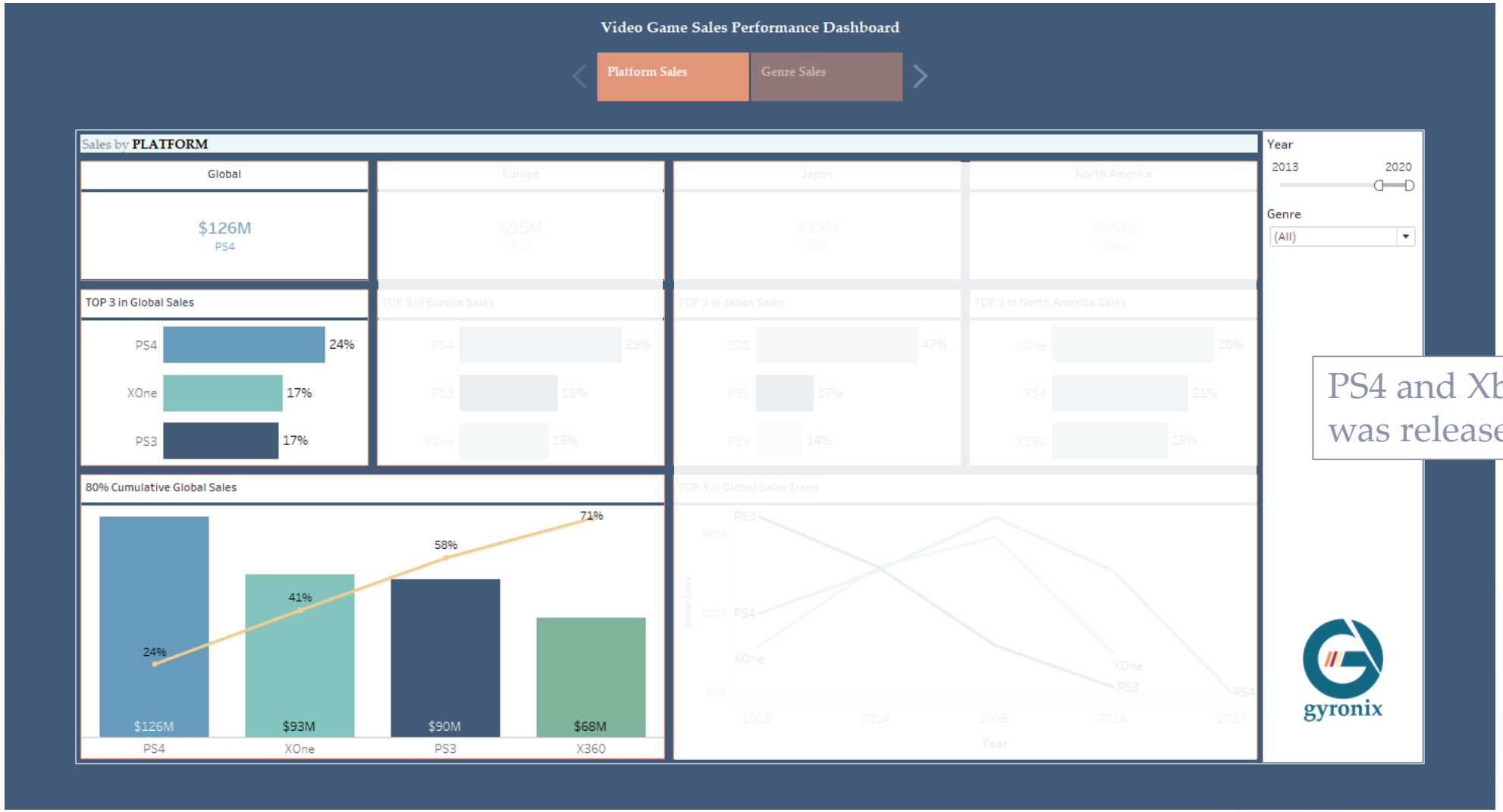
What is the **top performing genre** by global sales?

in Europe, Japan, and North America?

What **investment opportunities** can be found in the video game market?

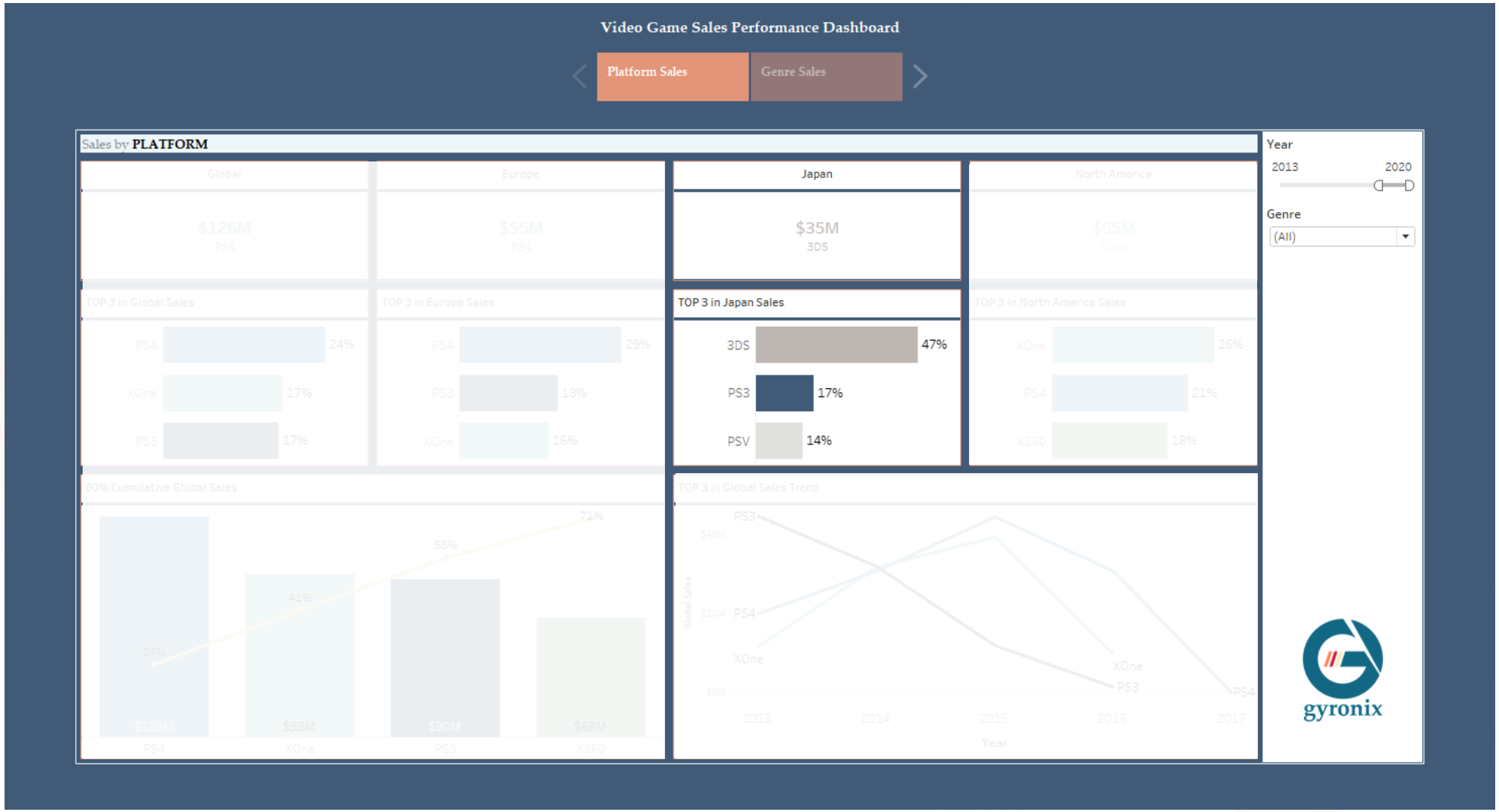
PLATFORM SALES

PS4 takes **24%** of the Global Sales

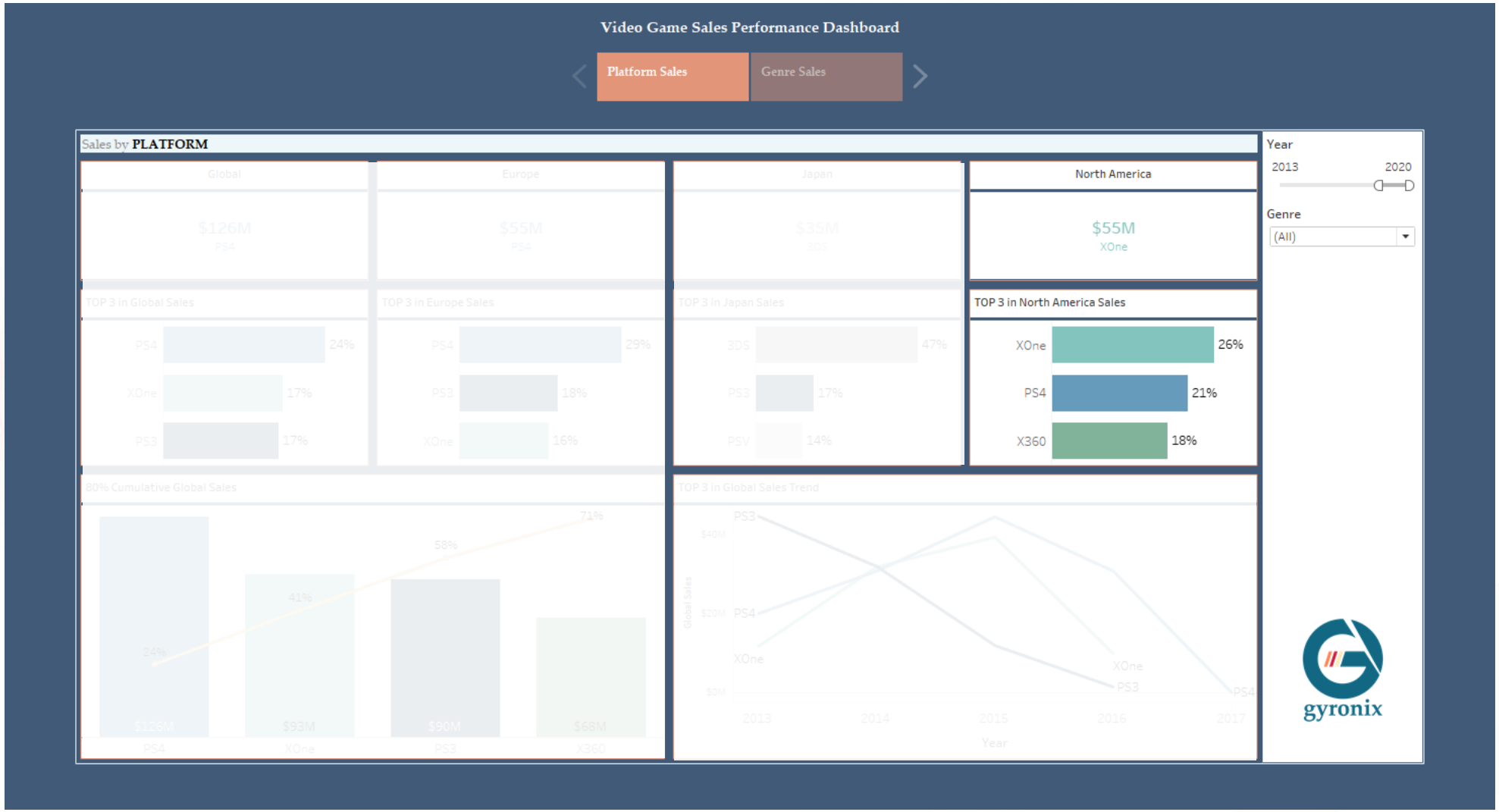


PS4 and Xbox One was released in **2013**

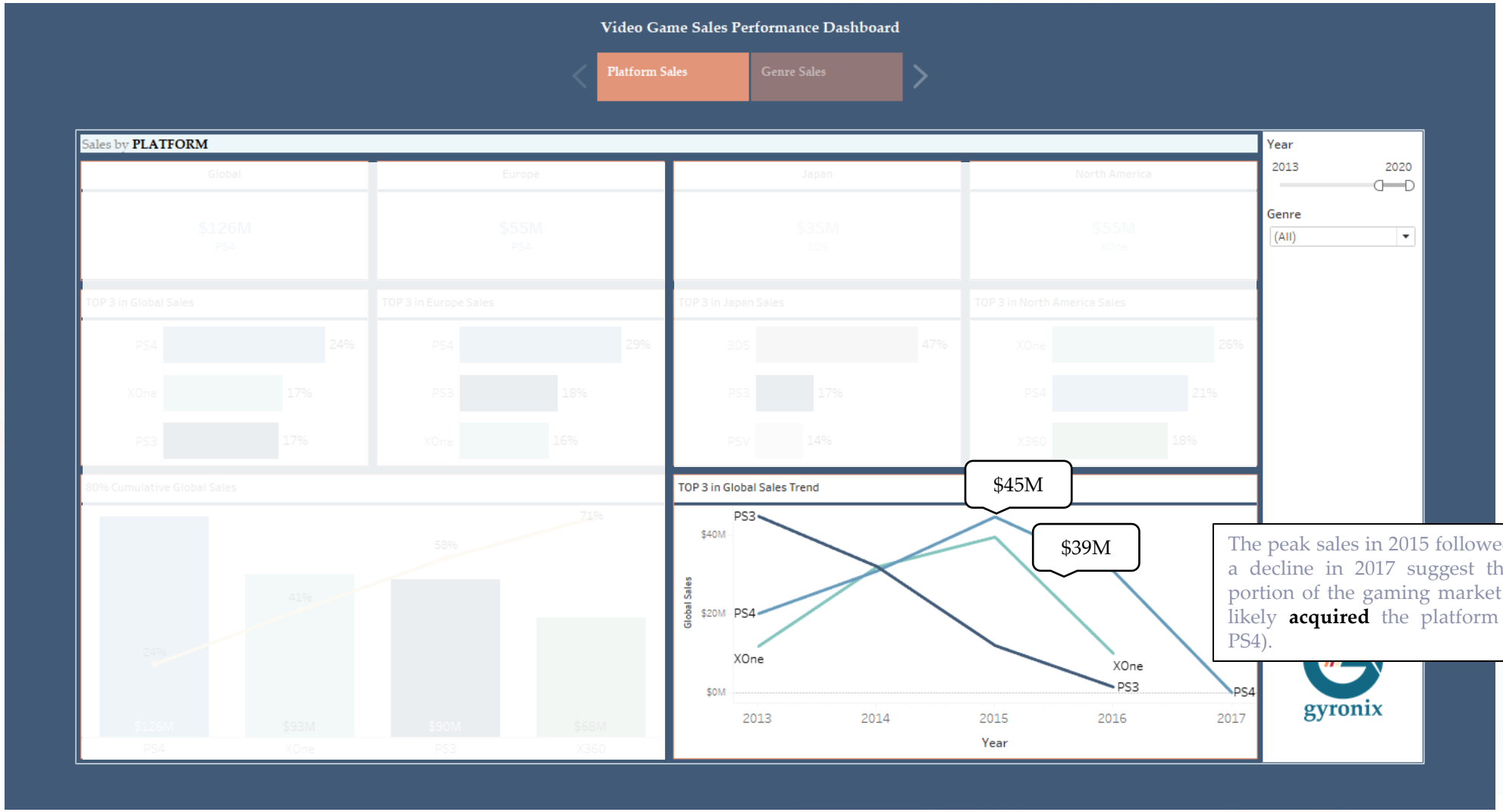
Japanese prefer **portable** gaming platform: Nintendo's **3DS**



Americans prefer **Xbox One**

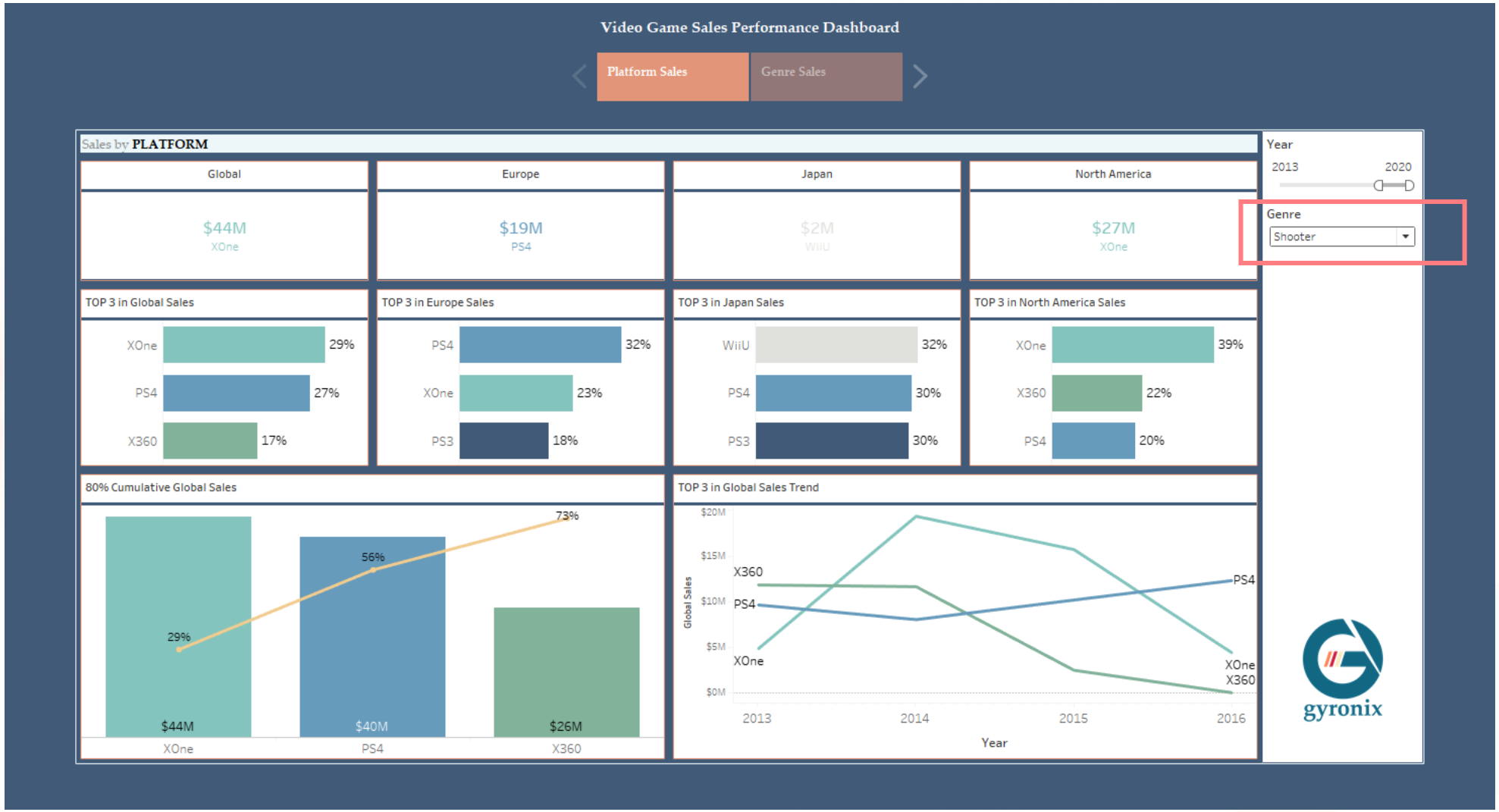


2 years after released (2015), **PS4** outperformed Xbox One by **$6M**



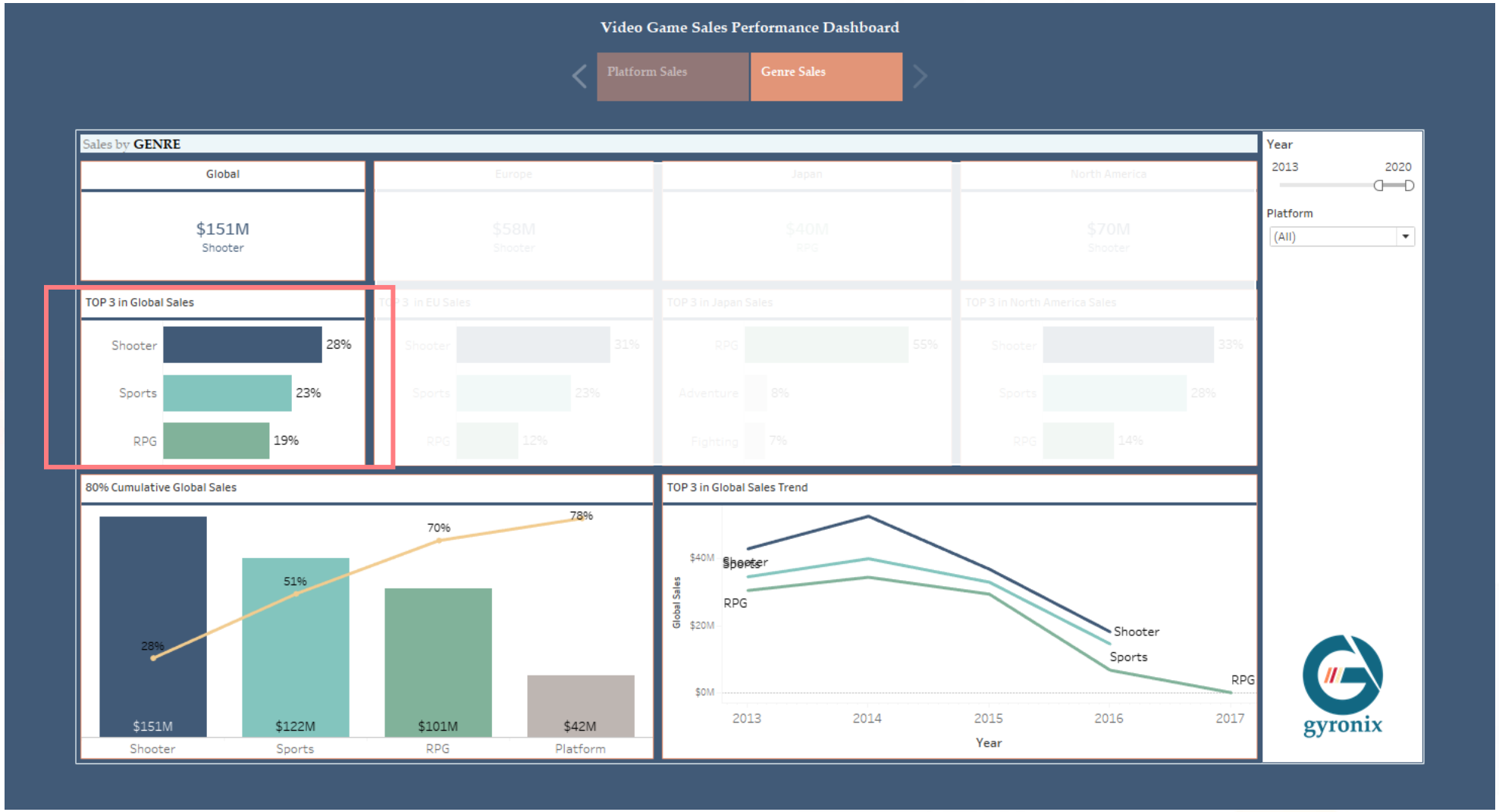
The peak sales in 2015 followed by a decline in 2017 suggest that a portion of the gaming market had likely **acquired** the platform (i.e. PS4).

Xbox One is the preferred platform for **Shooter** genre

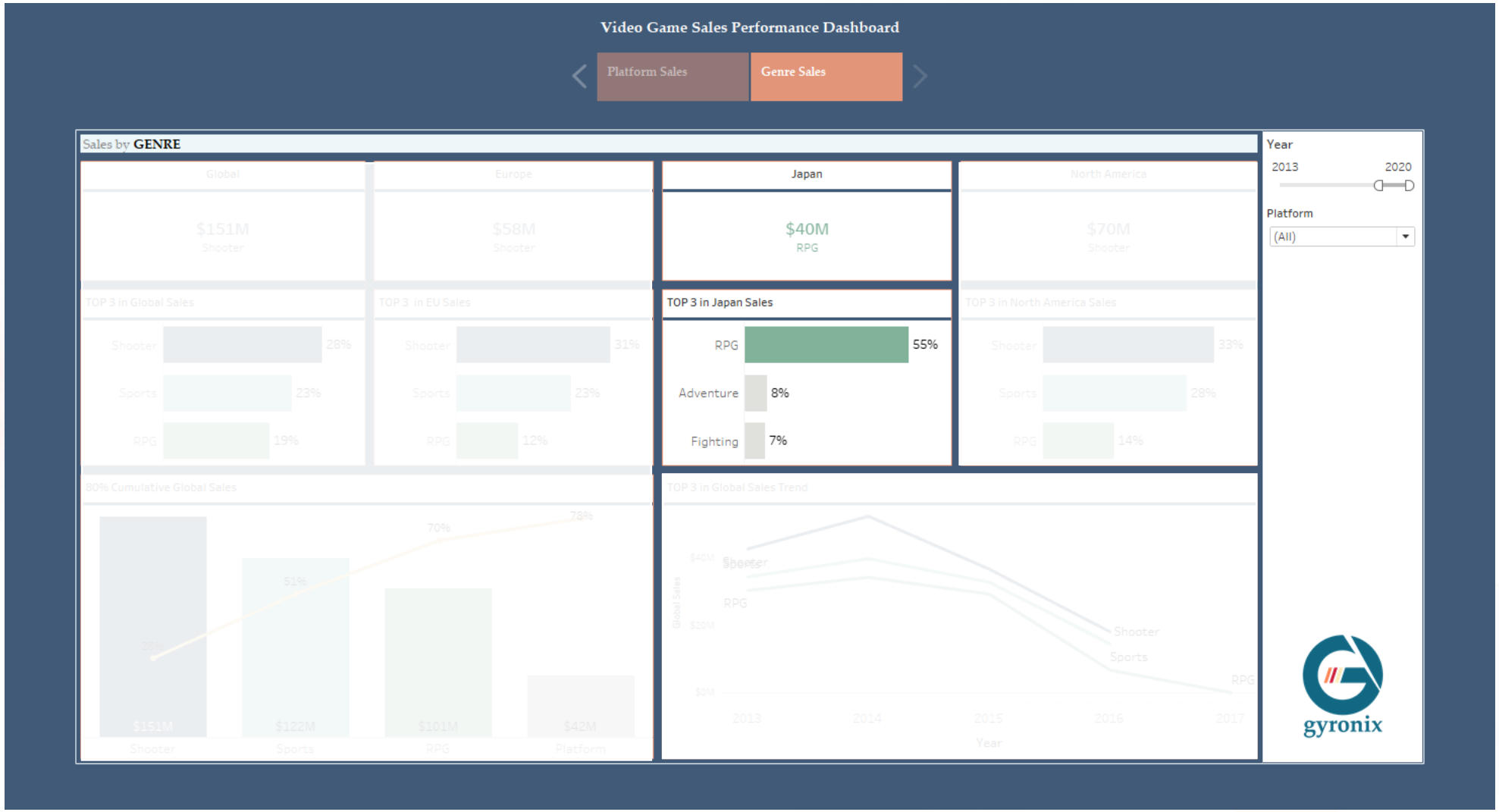


GENRE SALES

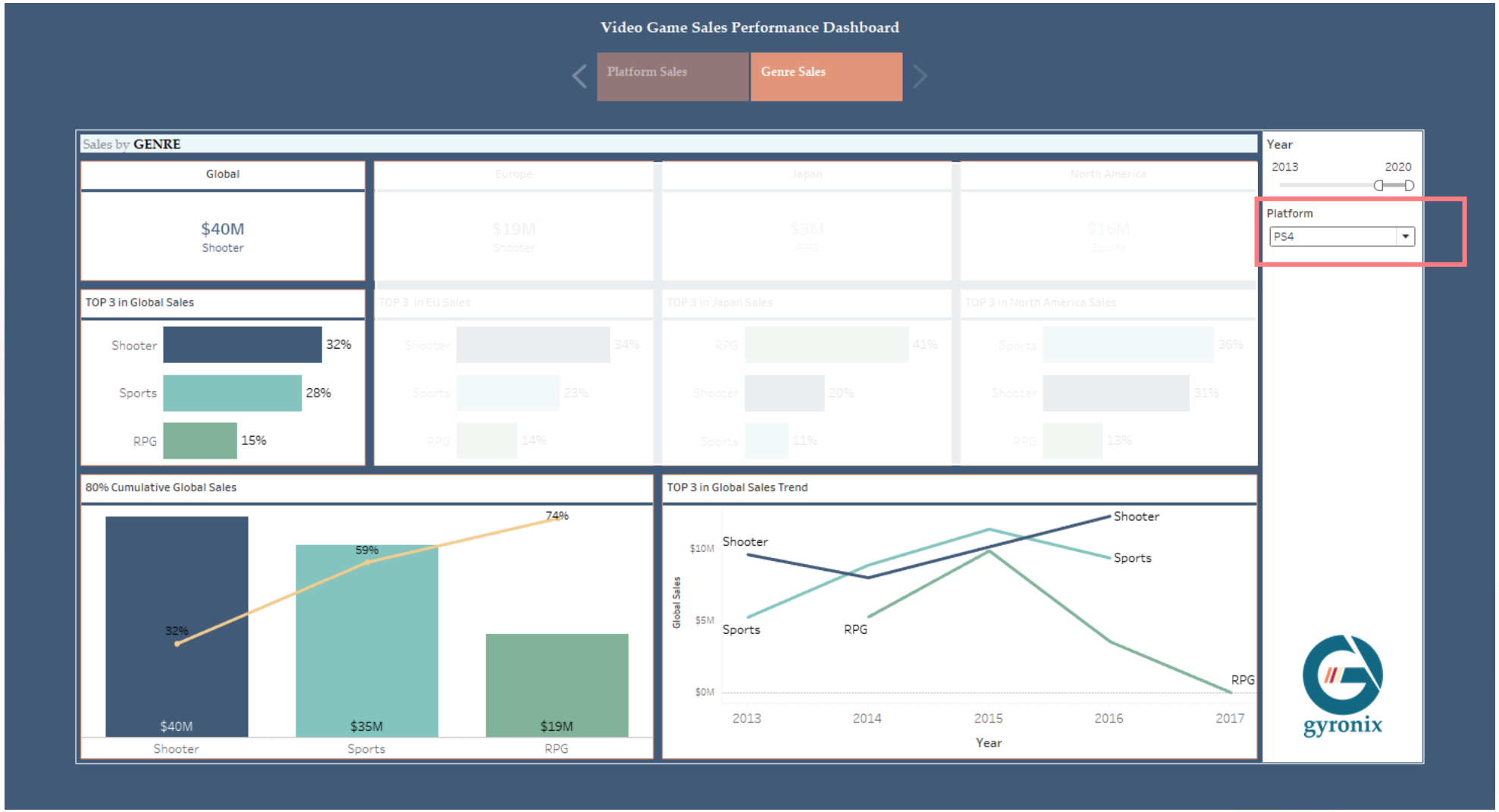
Shooter takes **28%** of the Global Sales



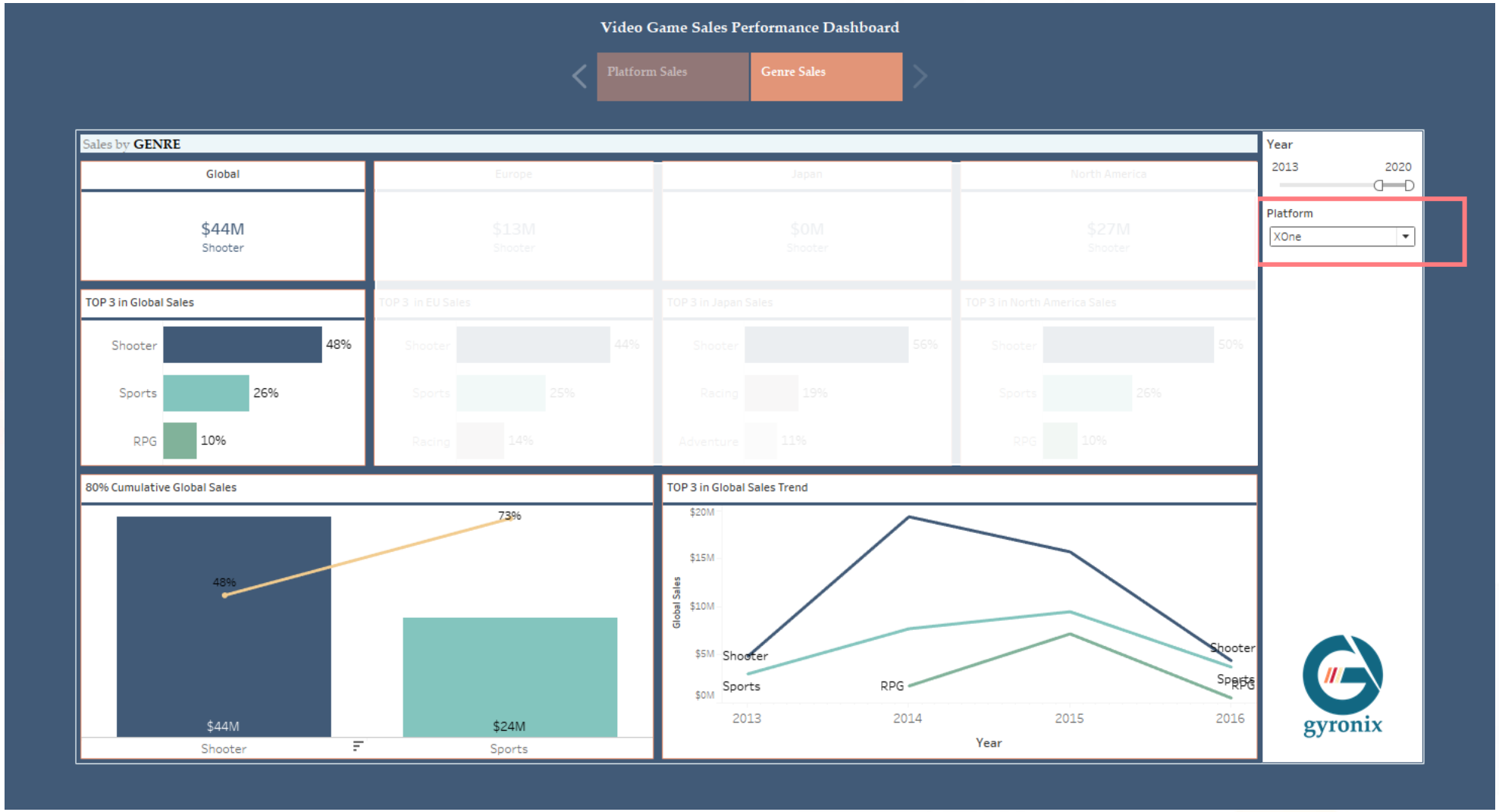
Japanese prefer RPG_



Shooter genre takes **32%** of Global **PS4** Sales



Shooter genre takes **48%** of Global **Xbox One** Sales



A/B TESTING

PS4 VS

XBOX ONE

HYPOTHESIS

PS4 Global Sales is **higher** than Xbox One

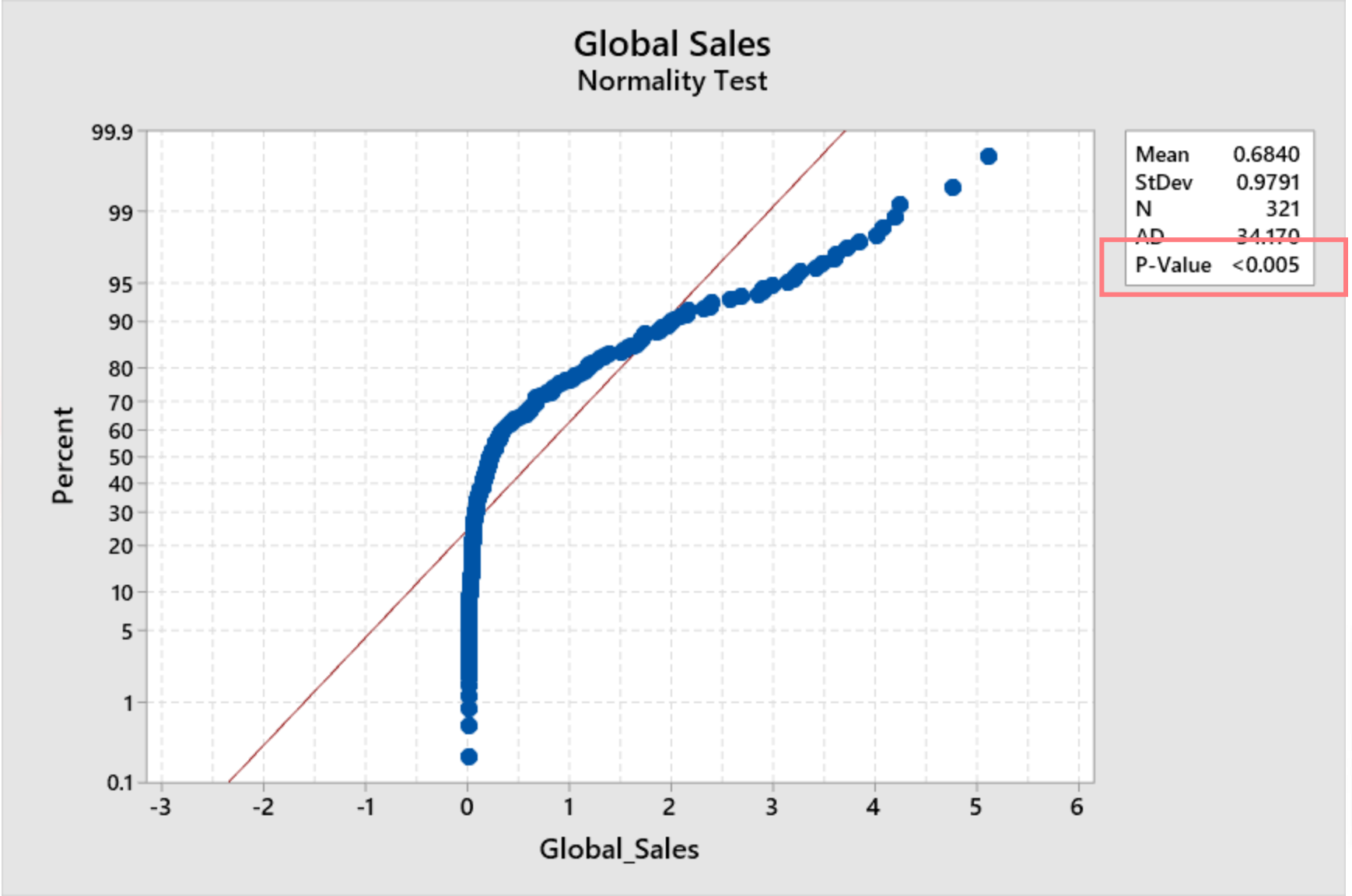
Null Hypothesis

H_o: **PS4 Sales = Xbox One Sales**

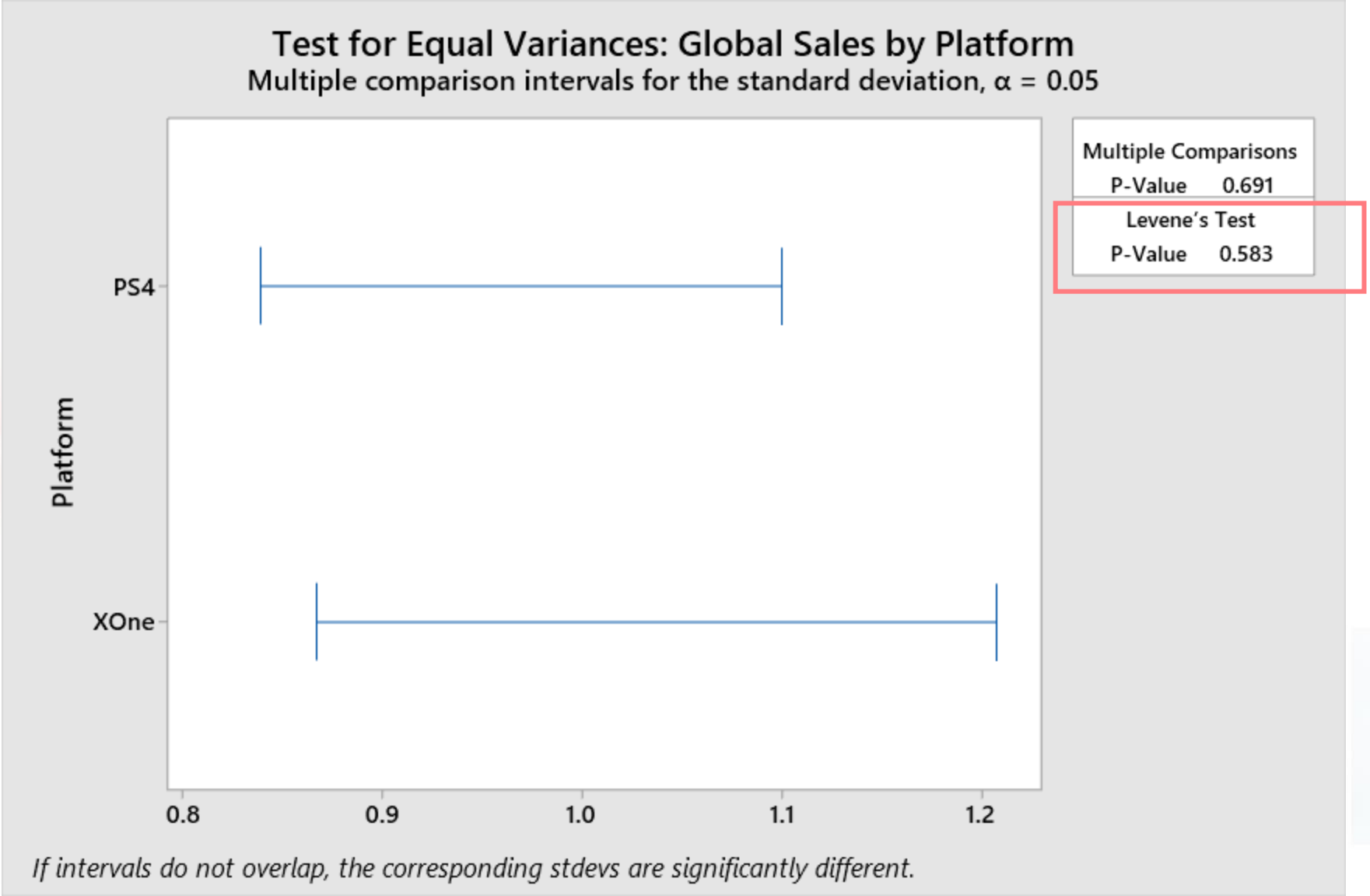
Alternative Hypothesis

H_a: **PS4 Sales > Xbox One Sales**

DATA IS NOT NORMAL



EQUAL VARIANCES



MANN-WHITNEY TEST

Descriptive Statistics

Sample	N	Median
Global_Sales_PS4	192	0.22
Global_Sales_XOne	129	0.25

Test

Null hypothesis	$H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis	$H_1: \eta_1 - \eta_2 > 0$

Method	W-Value	P-Value
Not adjusted for ties	30561.50	0.667

P-Value = 0.667 suggests that there is **no significant difference** in global sales between the PS4 and Xbox One platform

SHOOTER VS SPORTS

HYPOTHESIS

Null Hypothesis

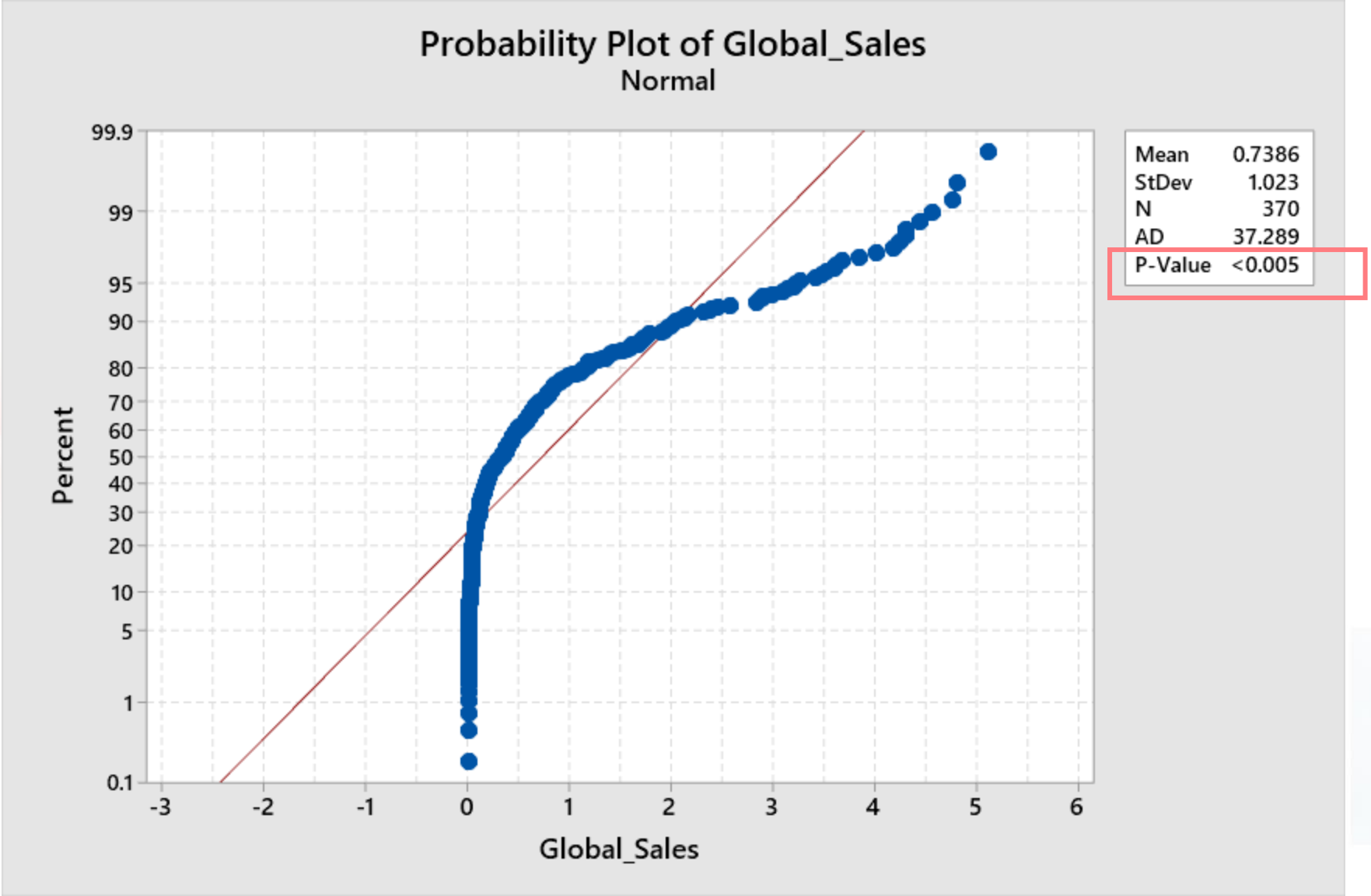
H_o: **Shooter Sales = Sports Sales**

Shooter global sales is **higher** than Sports

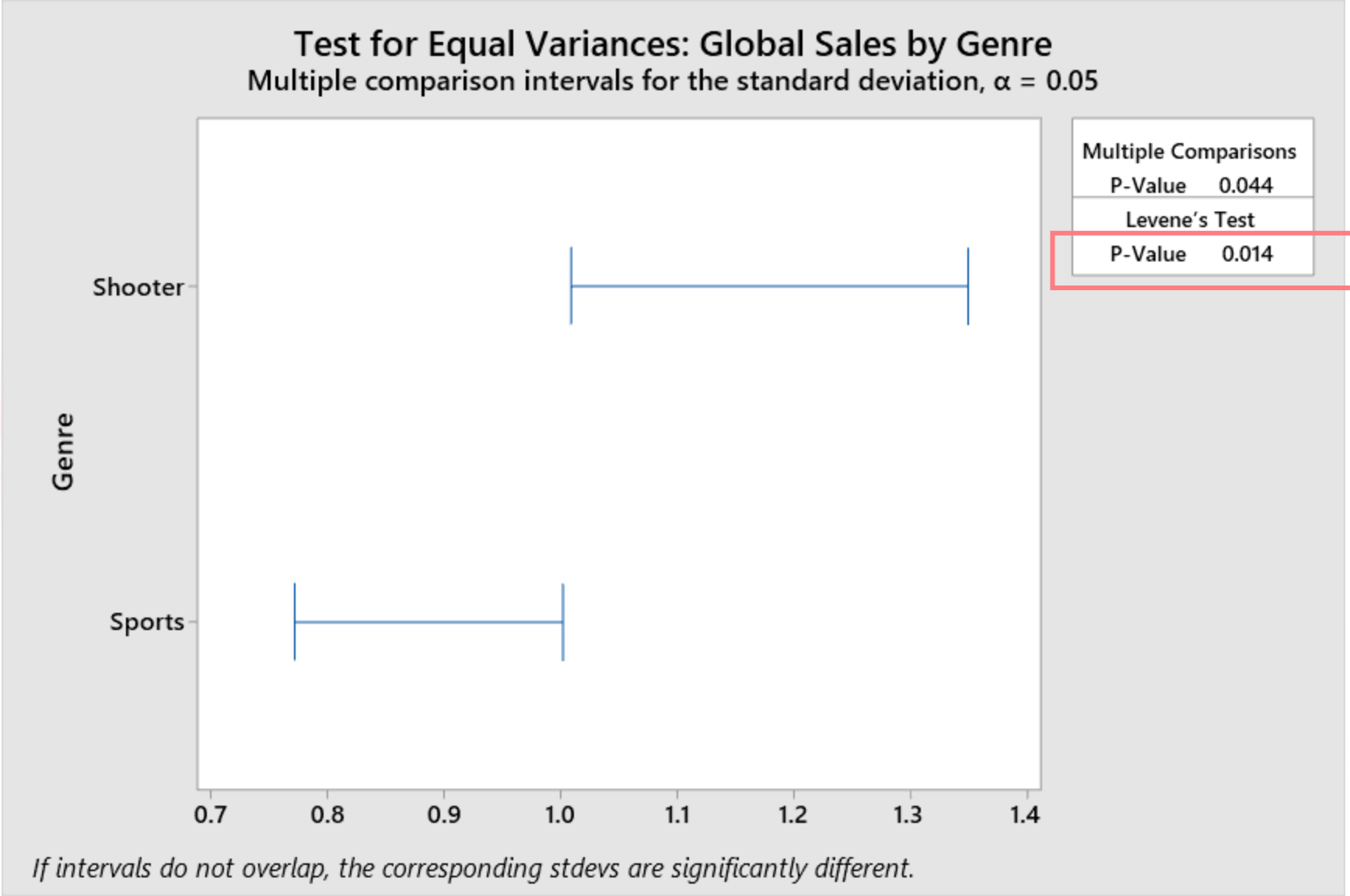
Null Hypothesis

H_a: **Shooter Sales > Sports Sales**

DATA IS NOT NORMAL



NOT EQUAL VARIANCES



MANN-WHITNEY TEST

Descriptive Statistics

Sample	N	Median
Global_Sales_Shooter	165	0.43
Global_Sales_Sports	205	0.20

Test

Null hypothesis	$H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis	$H_1: \eta_1 - \eta_2 > 0$

Method	W-Value	P-Value
Not adjusted for ties	33783.50	0.001
Adjusted for ties	33783.50	0.001

P-Value = 0.001 suggests that there is a **significant difference** in global sales between the Shooter and Sports genre

CONCLUSION

INVESTMENT OPPORTUNITY FOR SHOOTER GENRE

The hypothesis test results implies that, from a platform perspective, investing in either **PS4** or **Xbox One** may yield **similar returns** (p-value = 0.667). On the other hand, there appears to be a potential investment opportunity in the **shooter genre**, as it demonstrates **significantly higher** (p-value: 0.001) global sales compared to the sports genre.